

Database Systems Lab (CL1002)

Date: 2nd-December-2024

Course Instructor(s)

Ms. Fatima Gado

Lab Mid Exam (A)

Total Time: 120 minutes

Total Wtg: 50

Total Questions: 04

Semester: Fall-2024

Campus: Karachi

Department(s): CS

Submission Instructions:

- Return the question paper mention your **student ID on it**.
- Read each question completely before answering it. **There are 3 questions on 3 pages**.
- Attempt all the given questions. **All questions are carrying different points**.
- **Cheating in any case will lead to F-GRADE as per university rule.**
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.

Student Name

Roll No

Section

Student Signature

Question # 01: SQL Queries (LLO #: 3)

[5 Weightage]

1. Write a query to find the following output:

DEPARTMENT_NAME	EMPLOYEE_NAME	JOB_TITLE	SALARY
1 Accounting	Shelley Higgins	Accounting Manager	12008
2 Administration	Jennifer Whalen	Administration Assistant	4400
3 Executive	Steven King	President	24000
4 Finance	Nancy Greenberg	Finance Manager	12008
5 Human Resources	Susan Mavris	Human Resources Representative	6500
6 IT	Alexander Hunold	Programmer	9000
7 Marketing	Michael Hartstein	Marketing Manager	13000
8 Public Relations	Hermann Baer	Public Relations Representative	10000
9 Purchasing	Den Raphaely	Purchasing Manager	11000
10 Sales	John Russell	Sales Manager	14000
11 Shipping	Adam Fripp	Stock Manager	8200

- Find departments where the difference between the highest-paid and lowest-paid employee is greater than \$4000. Display the department name, maximum salary, minimum salary, and the salary difference.
- Identify the year and department that hired the most employees. Show the department name, year, and the number of hires.
- Find employees whose salary has not changed since they were hired. Display the employee name, hire date, current salary, and the department they work in.
- Write a query to find the employees who earn more than their direct manager. Display the employee name, manager's name, and their salary.

Note : Submission will be (Query + output screenshot).

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Question # 02: Triggers & Transaction (LLO #: 4)

[20 Weightage]

a. TRIGGER

15wtg

Product Table

product_id	product_name	price	stock_quantity
1	Laptop	50000.00	10
2	Smartphone	20000.00	8
3	Tablet	15000.00	20
4	Headphones	5000.00	6
5	Keyboard	2000.00	3

Using the provided data of Product Table, create a trigger named `stock_threshold_check` that fires **BEFORE UPDATE** on the Products table. The trigger should monitor the `stock_quantity` column during updates. If the new stock quantity (`NEW.stock_quantity`) is below the threshold value of 5, log a warning message in the `Stock_Alert` table.

`Stock_Alert` Table(alert_id, product_id, current_stock, alert_message & alert_date).

b. TRANSACTIONS

5wtg

Create a hotel guest reservation system. First, create the following tables: `guests` (`guest_id`, `name`, `email`), `rooms` (`room_id`, `room_type`, `price`), `reservations` (`reservation_id`, `guest_id`, `room_id`, `check_in`, `check_out`), `payments` (`payment_id`, `reservation_id`, `amount`), and `reservation_log` (`log_id`, `reservation_id`, `action`). Insert a new guest, select a room, make a reservation, process the payment, and log the action. Ensure that the transaction is committed only if all steps succeed; otherwise, roll back the changes.

c.

Question # 03: PL/SQL (LLO #: 4)

[15 Weightage]

Products Table

product_id	product_name	price	stock_quantity
1	Laptop	40000	10
2	Smartphone	25000	12
3	Headphones	500	8
4	Tablet	10000	15

Sales Table

sale_id	product_id	sale_date	sale_amount
1	1	2023-01-01	3
2	2	2023-01-15	5
3	3	2023-02-01	2

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1. A retail company maintains two tables: Products (columns: product_id, product_name, price, stock_quantity) and Sales (columns: sale_id, product_id, sale_date, sale_amount). Write a stored procedure named RecordSale that takes two input parameters (p_product_id and p_sale_amount). The procedure should: Check if the product exists in the Products table. If not, display a message: "Product not found." If the product is found, check if the sale amount is greater than the stock quantity. If yes, display a message: "Insufficient stock for the sale." If the sale is possible, update the stock quantity in the Products table and insert a record in the Sales table.
2. Write a stored function named GetTotalSalesAmount that takes a product_id as an input parameter and returns the total sales amount for that product by summing the sale amounts from the Sales table.

Question # 04: MongoDB (LLO #: 3)**[10 Weightage]**

- Insert the following entries in mongoDB.

```
[
  { "book_id": "B001", "title": "Data Science Fundamentals", "author": "John Smith", "category":
    "Technology", "price": 29.99, "in_stock": true },
  { "book_id": "B002", "title": "Learning MongoDB", "author": "Jane Doe", "category": "Technology",
    "price": 35.99, "in_stock": false },
  { "book_id": "B003", "title": "Web Development with JavaScript", "author": "Mark Lee", "category":
    "Programming", "price": 24.99, "in_stock": true },
  { "book_id": "B004", "title": "Introduction to Python", "author": "Alice Brown", "category":
    "Programming", "price": 19.99, "in_stock": true },
  { "book_id": "B005", "title": "Advanced SQL Queries", "author": "Michael White", "category":
    "Database", "price": 45.99, "in_stock": true },
  { "book_id": "B006", "title": "C++ Basics", "author": "John Smith", "category": "Programming",
    "price": 29.99, "in_stock": true },
  { "book_id": "B007", "title": "Machine Learning with Python", "author": "Sara Green", "category":
    "Technology", "price": 39.99, "in_stock": true },
  { "book_id": "B008", "title": "Deep Learning Essentials", "author": "David Grey", "category":
    "Technology", "price": 59.99, "in_stock": false },
  { "book_id": "B009", "title": "Data Structures in Java", "author": "Lucas Blue", "category":
    "Programming", "price": 49.99, "in_stock": true },
  { "book_id": "B010", "title": "Artificial Intelligence", "author": "Sophia Grey", "category":
    "Technology", "price": 69.99, "in_stock": true }
]
```

1. Update the price of all books in the "Technology" category that have a price less than \$40, and set the in_stock value to false.
2. Delete a book where the book_id is "B008" and it is out of stock (in_stock: false).
3. Update the price of books by "John Smith" in the "Technology" category to \$35, but only if the price is greater than \$20 and the book is in stock.
4. Find all books from the "Programming" category with a price greater than \$25, and where the in_stock value is true. Sort them by price in descending order.
5. Delete all books that are either in the "Programming" category and out of stock (in_stock: false), or have a price greater than \$50.
6. Update the price of books with book_id "B005" and "B006" to \$40, and set their in_stock value to true.

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7. Find all books where the author is "Jane Doe" or the category is "Database", and the price is between \$30 and \$60.
8. Update the price of all books in the "Programming" category to \$50, only if their current price is less than \$50 and they are currently in stock.
9. Delete all books where the price is less than \$30 and the category is "Technology", and their in_stock value is false.
10. Update the title of the book with book_id "B010" to "AI Revolution", and reduce its price by 10%.

Note : Submission will be (Query + output screenshot).

BEST OF LUCK☺